

GALAXY EVOLUTION

22–23 The scale of the universe

54–55 Out of the darkness

302–305 Types of galaxy

Active galaxies 320–321

Galaxy clusters 326–327

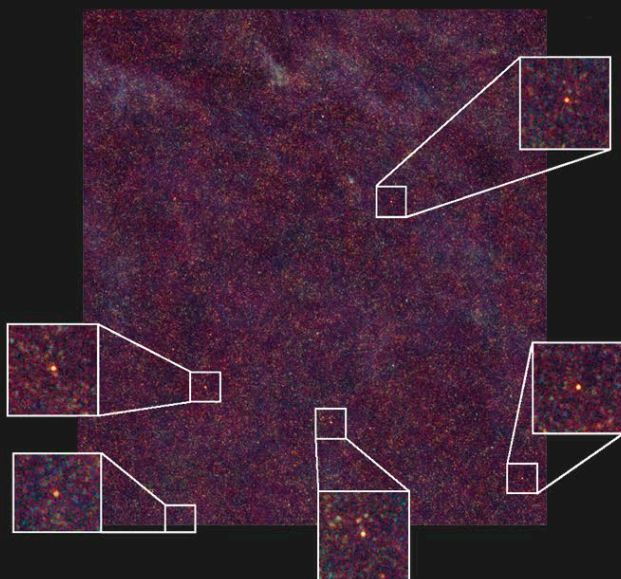
Galaxy superclusters 336–339

THE PROCESSES BY WHICH DIFFERENT types of galaxies form have puzzled astronomers for almost a century, but today, a new generation of telescopes capable of studying galaxies billions of light-years away is finally resolving some

key questions. Light from these remote galaxies left on its long journey to Earth when the universe was very young, so it can reveal the secrets of the early stages of galactic evolution.

THE DISTRIBUTION OF GALAXIES

Astronomers can only ever see a “snapshot” of a brief moment in a galaxy’s long life story, so they have to build up a picture of galactic evolution by studying many individual galaxies. Such studies have revealed certain patterns, such as the fact that large elliptical galaxies are found only in substantial galaxy clusters. Changes in the type of galaxies seen at different distances—and therefore at different stages in cosmic history—can also reveal patterns in the way galaxies have developed. However, capturing the light of the most distant early galaxies is an enormous challenge, requiring techniques such as long-exposure deep-field photography and the use of gravitational lensing.



GRAVITATIONALLY LENSED GALAXIES

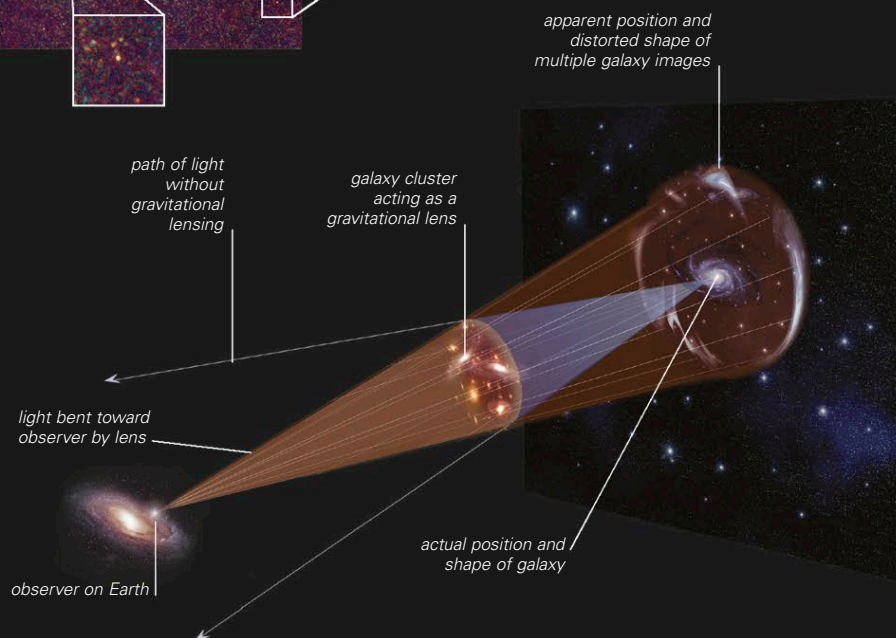
This image of a small area in the constellation Hydra was taken by the Herschel Space Observatory and reveals more than 6,000 galaxies. The white squares indicate distant galaxies that have been gravitationally lensed by foreground galaxies. These distant galaxies are brighter at the submillimeter wavelengths detected by Herschel than at visible-light wavelengths.

DUSTY LENTICULAR GALAXY

This image of the lenticular galaxy NGC 1316 in the constellation of Fornax was taken by the Hubble Space Telescope. It reveals a complex series of dust lanes and dust patches in the galaxy, indicating that it was formed from the merger of two galaxies rich in dust and gas.

GRAVITATIONAL LENSING

The technique of gravitational lensing relies on the fact that the gravity of a massive object deflects light passing nearby—in effect, the object acts as a lens. In this way, a massive object, such as a galaxy cluster, situated between Earth and a more distant galaxy (or other object) can focus light from the distant galaxy onto Earth. This often produces multiple distorted images of the distant galaxy but also intensifies its light.



EXPLORING SPACE

HERSCHEL

Operational from 2009 to 2013, the European Space Agency’s Herschel Space Observatory was designed to observe the longest infrared wavelengths (the far infrared and submillimeter wavelengths on the boundary with radio waves). Its primary mirror was 11.5 ft (3.5 m) in diameter and its instruments were cooled to -456°F (-271°C), enabling it to map some of the coolest and most distant objects in the universe.



THE HERSCHEL SPACE OBSERVATORY